Meta-Learning for Crisis-Aware Portfolio Allocation: A Jump Model Approach

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***Abstract*—Financial crises pose significant challenges for portfolio allocation models due to their rarity, unique characteristics, and the need for rapid adaptation. Traditional machine learning approaches struggle during market disruptions because of data scarcity and distribution shifts. We present a novel application of Model-Agnostic Meta-Learning (MAML) combined with statistical jump detection for crisis-aware portfolio allocation. Our approach uses the Lee-Mykland jump model to detect regime changes and constructs meta-learning tasks from historical market periods, enabling rapid adaptation to emerging crises with limited data. Trained on 35 years of S&P 500 data spanning seven major crises, our model achieves a test loss of 0.0251 (15.8% RMSE) and demonstrates strong out-of-distribution generalization on the COVID-19 crisis with only 1.87% prediction bias, representing a 91% improvement in bias reduction compared to baseline approaches. Portfolio backtesting validates practical effectiveness with a Sharpe ratio of 0.47, outperforming buy-and-hold (0.36) and traditional 60/40 allocation (0.32). Our results demonstrate that pattern diversity in training tasks is more critical than model complexity for small-data meta-learning regimes.**

***Index Terms*—meta-learning, MAML, portfolio allocation, crisis prediction, regime detection, jump models, financial machine learning**

I. Introduction

Financial crises represent critical challenges for portfolio management systems. Market disruptions such as the 2008 financial crisis, 2020 COVID-19 pandemic, and 2022 Ukraine war create rapid shifts in asset behavior that traditional models fail to capture effectively. Three fundamental problems emerge: (1) data scarcity, with only 5-10 major crises occurring over 30 years; (2) distribution shift, where each crisis exhibits unique characteristics; and (3) adaptation speed, requiring models to adjust quickly as crises unfold.

Traditional machine learning approaches require substantial training data and struggle with out-of-distribution scenarios. Regime-switching models like Hidden Markov Models (HMMs) rely on predefined states and often exhibit lag in detecting transitions. Deep learning methods demand extensive retraining when market conditions change. These limitations motivate our investigation of meta-learning for crisis-aware portfolio allocation.

Meta-learning, or "learning to learn," offers a solution by training models to adapt quickly to new tasks with minimal data. Model-Agnostic Meta-Learning (MAML) [1] has shown success in few-shot learning across computer vision and natural language processing. We apply MAML to financial crisis prediction, treating each historical market regime as a distinct task. Our model learns patterns across multiple crises, enabling rapid adaptation when new disruptions emerge.

Our key contributions are:

- A novel application of MAML to crisis-aware portfolio allocation using jump model regime detection
- Demonstration that pattern diversity in training tasks exceeds model complexity in importance for small-data meta-learning
- Strong out-of-distribution generalization validated on COVID-19 crisis (1.87% bias)
- Empirical validation through portfolio backtesting achieving 0.47 Sharpe ratio
- Competitive performance with 0.0251 test loss (15.8% RMSE) on our 35-year dataset

II. Related Work

A. Regime Detection in Financial Markets

Regime detection methods aim to identify distinct market states. HMMs represent the classical approach, modeling market dynamics as transitions between hidden states [16]. However, HMMs suffer from lag and require careful threshold calibration. The Volatility Index (VIX) serves as a popular fear indicator but spikes reactively after crises begin rather than predictively.

Jump models provide an alternative framework based on statistical detection of discontinuous price movements. Lee and Mykland [12] developed a nonparametric test using bipower variation for robust volatility estimation. This approach detects actual market movements rather than sentiment, reducing lag compared to VIX-based methods.

B. Meta-Learning

MAML [1] optimizes for rapid adaptation by training a model initialization that enables fast fine-tuning on new tasks. The algorithm maintains gradient flow through the adaptation process using higher-order derivatives. Applications span computer vision [41], robotics, and natural language processing.

Financial applications of meta-learning remain limited. Recent work explores few-shot learning for stock prediction [9] and portfolio optimization [7], but focuses on individual assets rather than crisis regimes. Our work applies MAML specifically to crisis-aware portfolio allocation with jump-based regime detection.

C. Crisis Prediction and Portfolio Allocation

Traditional crisis prediction relies on early warning indicators such as credit spreads, yield curves, and volatility measures [27]. Machine learning approaches include random forests and recurrent neural networks [21]. These methods require retraining for new crises and lack fast adaptation mechanisms.

Portfolio optimization has incorporated machine learning techniques [18] and reinforcement learning [17], but adapting to sudden regime changes remains challenging. Our approach addresses this gap through meta-learning.

III. Methodology

A. Data and Features

We utilize 35 years of daily S&P 500 and VIX data (1990-2025, 8,970 trading days) from Yahoo Finance. Our feature engineering generates 32 indicators across five categories:

Returns and Momentum (7 features): Log returns, moving averages (5-day, 20-day), and momentum indicators capture price trends and short-term dynamics.

Volatility Indicators (5 features): Rolling standard deviation, Average True Range (ATR), VIX changes, and spike indicators measure market turbulence.

Technical Indicators (4 features): Relative Strength Index (RSI), volume changes, and high-low ranges provide additional context.

VIX Thresholds (4 features): Binary indicators for VIX levels above 20, 30, and 40 capture fear regime transitions.

Jump Model Features (12 features): Jump detection indicators, jump intensity over 5-day and 20-day windows, cluster metrics, average jump sizes, and negative jump ratios quantify discontinuous market behavior.

All features undergo standardization using zero-mean, unit-variance scaling to ensure stable gradient flow during training.

B. Jump-Based Regime Detection

Our regime classification builds on the Lee-Mykland jump detection framework [12]. For each day t, we compute the test statistic:

$$L_t = \frac{r_t - \mu}{\sigma_t} \tag{1}$$

where r_t represents the log return, μ denotes the local return mean, and σ_t indicates bipower variation providing robust volatility estimation:

$$BV_t = \sum_{i=1}^{n-1} |r_i| \times |r_{i-1}| \times \frac{\pi}{2} \tag{2}$$

We detect jumps when $|L_t|$ exceeds a threshold derived from the inverse cumulative distribution function with 1% significance level. Jump classification includes binary detection, magnitude, and direction.

Using rolling 20-day windows, we compute jump frequency and average jump size to assign regime labels:

- **JR0 (Calm):** Jump frequency $< 10\%$ and average size $< 2\%$
- **JR1 (Moderate):** $10\% \leq$ frequency $< 20\%$ or size $< 3\%$
- **JR2 (Crisis):** Frequency $\geq 20\%$ or size $\geq 3\%$

This approach captures seven major crises including the 1998 Asian crisis, 2002 WorldCom collapse, 2008 financial crisis, 2009 recovery, 2011 debt downgrade, 2020 COVID-19 pandemic, and 2025 market volatility, achieving superior coverage compared to VIX-only methods.

C. Task Generation

Meta-learning requires multiple training tasks. We segment the time series at regime change points, growing segments while maintaining dominant regime purity above 60%. Tasks span 20-90 days, ensuring sufficient data for support-query splits while capturing crisis diversity.

Each task contains:

- **Support set:** 20 samples for adaptation
- **Query set:** 10 samples for evaluation
- **Metadata:** Regime label, purity, jump statistics

Our final dataset comprises 69 tasks: 29 JR0 (calm), 15 JR1 (moderate), and 7 JR2 (crisis). The temporal split ensures no data leakage: 35 training tasks (including 3 JR2), 8 validation tasks (2 JR2), and 26 test tasks (2 JR2 including COVID-19 as out-of-distribution holdout).

D. MAML Training

We implement MAML using the learn2learn library [6] to ensure proper gradient flow through adaptation. Our architecture consists of a 2-layer multilayer perceptron with 64 hidden units predicting next-day log returns.

The training process alternates between inner and outer loops:

Inner Loop (Adaptation): For each task, we clone the meta-model and perform 5 gradient descent steps on the support set:

$$\theta' = \theta - \alpha \nabla_\theta \mathcal{L}_{support}(\theta) \tag{3}$$

where $\alpha = 0.01$ represents the inner learning rate.

Outer Loop (Meta-Update): We evaluate adapted parameters on query sets and compute the meta-loss:

$$\mathcal{L}meta = \frac{1}{|B|} \sum i \in B \mathcal{L}_{query}(\theta_i') \tag{4}$$

where B denotes the task batch. The meta-optimizer (Adam with learning rate 0.001) updates the base parameters to minimize query loss across all tasks:

$$\theta \leftarrow \theta - \beta \nabla_\theta \mathcal{L}_{meta} \tag{5}$$

This procedure enables the model to learn an initialization that adapts quickly to new market regimes with minimal data.

IV. Experiments and Results

A. Training Configuration

We train for 50 epochs with batch size 4 (tasks per batch) on Tesla T4 GPU hardware. Training completes in 12 minutes, demonstrating computational efficiency. We apply gradient clipping (max norm 5.0) to both inner and outer loops for stability.

B. Overall Performance

Table I presents our main results. The model achieves 0.0251 test loss (15.8% RMSE) across all regimes, with strong performance in calm markets (15.3% RMSE) and moderate conditions (14.7% RMSE). Crisis prediction reaches 23.6% RMSE, representing competitive performance on our dataset.

TABLE I
Test Performance by Regime

Regime	Loss (MSE)	RMSE (%)	Tasks
JR0 (Calm)	0.0234	15.3	18
JR1 (Moderate)	0.0215	14.7	6
JR2 (Crisis)	0.0557	23.6	2
Overall	**0.0251**	**15.8**	**26**

C. Crisis Task Analysis

Table II details performance on all seven crisis tasks. Our model demonstrates strong generalization on COVID-19 (test set) with only 1.87% positive bias despite never observing similar V-shaped recoveries during training. The 2011 debt crisis shows exceptional accuracy (1.48% bias), while the 2008 financial crisis remains challenging with 24.8% positive bias reflecting extreme severity.

TABLE II
Crisis Task Performance Analysis

Event	Period	Split	Loss	Bias
1998 Asian	Aug 31 - Sep 30 (21d)	Train	0.0046	+2.12%
2002 WorldCom	Jul 23 - Aug 22 (22d)	Train	0.0073	+2.32%
2008 Financial	Sep 18, 2008 - Jan 6, 2009 (75d)	Train	0.0735	+24.8%
2009 Recovery	Feb 24 - Apr 7, 2009 (30d)	Val	0.0151	+5.63%
2011 Debt	Aug 9 - Sep 8 (21d)	Val	0.0084	+1.48%
2020 COVID	Mar 4 - Apr 24 (36d)	Test	0.0111	+1.87%
2025 Recent	Apr 7 - May 6 (20d)	Test	0.0358	-4.44%
Average	–	–	**0.0223**	**+4.83%**

D. Portfolio Backtesting

We validate practical effectiveness through portfolio backtesting on the same 35-year period. Using MAML predictions for regime-aware allocation (60% equity in calm markets, 50% in moderate volatility, 30% in crises), our strategy achieves:

- Sharpe Ratio: 0.47 (vs. Buy-and-Hold 0.36, 60/40 0.32, 40/60 0.24)
- Cumulative Return: 20.5x (vs. Buy-and-Hold 14.0x)
- Annualized Return: 8.9% with 14% volatility
- Maximum Drawdown: -51.2% (similar to baselines but faster recovery)

These results demonstrate that meta-learned crisis predictions translate to superior risk-adjusted portfolio returns.

E. Ablation Studies

We conducted extensive ablation studies to isolate key factors:

Task Threshold Impact: Lowering the minimum task duration from 30 to 20 days increased crisis task count from 5 to 7, improving test loss by 35.9% and crisis loss by 69.7%. This demonstrates that pattern diversity dominates model complexity.

Gradient Flow: Manual MAML implementation using copy.deepcopy() failed catastrophically, while learn2learn with proper higher-order gradients succeeded immediately. This validates the critical importance of maintaining computational graphs through adaptation.

Jump Model vs. VIX: Our jump-based regime detection captured the 2022 Ukraine war and 2002 WorldCom crisis missed by VIX-only approaches, contributing to improved coverage.

Bias Evolution: Comparison with baseline approach (30-day threshold, 5 crisis tasks) shows significant improvement: COVID-19 bias reduced from -22.5% (defensive) to +1.87% (nearly neutral), representing 91% bias reduction.

F. Comparison with Baselines

Table III compares our approach with baseline models on our dataset. MAML demonstrates superior crisis handling compared to naive mean prediction, autoregressive models, and demonstrates competitive performance with random forests and LSTM networks. The fast adaptation capability (approximately 100ms with 20 samples) enables real-time deployment.

TABLE III
Comparison with Baseline Models on Our Dataset

Model	Test Loss	JR2 Loss
MAML (Ours)	**0.0251**	**0.0557**
Naive Mean	0.130	0.280
AR(1)	0.055	0.220
Random Forest	0.048	0.210
LSTM	0.042	0.195

V. Discussion

Our results validate meta-learning for crisis-aware portfolio allocation. The strong COVID-19 generalization (1.87% bias) despite out-of-distribution characteristics demonstrates that MAML learns transferable crisis patterns rather than memorizing historical events.

The critical role of pattern diversity emerged as our key finding. Adding two diverse crises (2002 sharp correction and 2025 volatility) to training yielded greater improvement than any architectural modification. This insight guides future work: expanding to international markets to capture 15-20 crisis tasks should further enhance generalization.

The jump model framework proved superior to VIX-based detection by capturing actual price movements rather than sentiment. Jump frequency and magnitude provide leading indicators that reduce lag compared to reactive fear measures.

Limitations include the 2008 outlier (24.8% bias) indicating extreme events remain challenging, and the requirement for 7+ diverse crisis tasks limiting applicability to short historical datasets. The portfolio backtest assumes zero transaction costs and perfect execution, which may overestimate real-world performance.

Future work should explore: (1) international market expansion for diverse crisis patterns, (2) hierarchical MAML for crisis subtypes, (3) incorporation of transaction costs in portfolio optimization, and (4) real-time deployment and monitoring systems.

VI. Conclusion

We presented a novel application of meta-learning combining MAML with jump model regime detection for crisis-aware portfolio allocation. Our approach achieves competitive crisis prediction accuracy (23.6% RMSE on crisis regimes) and strong out-of-distribution generalization (1.87% COVID-19 bias) while adapting rapidly to new crises with only 20 samples and 5 gradient steps.

The validation that pattern diversity exceeds model complexity in importance for small-data meta-learning provides actionable guidance for practitioners: invest in diverse training scenarios rather than architectural sophistication when working with limited crisis data.

Portfolio backtesting demonstrates practical effectiveness with 0.47 Sharpe ratio, outperforming traditional strategies on our 35-year dataset. The combination of statistical jump detection and meta-learning represents a principled approach to the fundamental challenges of crisis prediction: data scarcity, distribution shift, and adaptation speed.

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References

[1] C. Finn, P. Abbeel, and S. Levine, "Model-agnostic meta-learning for fast adaptation of deep networks," in *Proc. 34th Int. Conf. Mach. Learn. (ICML)*, 2017, pp. 1126–1135.
[2] L. Collins et al., "Task-robust model-agnostic meta-learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
[3] J. Wang et al., "Towards task sampler learning for meta-learning," *arXiv preprint arXiv:2307.08924*, 2023.
[4] O. Bohdal et al., "EvoGrad: Efficient gradient-based meta-learning and hyperparameter optimization," *OpenReview*, 2021.
[5] Y. Park et al., "Meta-learning with adaptive hyperparameters," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
[6] A. Beaulieu et al., "learn2learn: A library for meta-learning research," 2020. [Online]. Available: https://github.com/learnables/learn2learn
[7] "Meta-learning the optimal mixture of strategies for online portfolio selection," *arXiv preprint*, 2024.
[8] T. Liu et al., "A stock price prediction method based on meta-learning and variational mode decomposition," *Knowledge-Based Systems*, 2022.
[9] "Few-shot learning patterns in financial time-series for trend-following strategies," *arXiv preprint arXiv:2310.10500*, 2020.
[10] "Meta-learning framework for effective few shot time series forecasting," *IEEE Trans. Knowledge Data Eng.*, vol. 38, no. 1, Jan. 2025.
[11] "Meta-learning for few-shot time series forecasting," in *Proc. ACM Int. Conf. Information Knowledge Management*, 2021.
[12] S. S. Lee and P. A. Mykland, "Jumps in financial markets: A new nonparametric test and jump dynamics," *Rev. Financial Studies*, vol. 21, no. 6, pp. 2535–2563, 2008.
[13] O. E. Barndorff-Nielsen and N. Shephard, "Power and bipower variation with stochastic volatility and jumps," *J. Financial Econometrics*, vol. 2, no. 1, pp. 1–37, 2004.
[14] K. Korangi and K. Khashanah, "A new method for jump detection: Analysis of jumps in the S&P 500 index," *J. Royal Statistical Society Series C*, 2025.
[15] "Regime switching models in finance and economics," *Int. J. For Multidisciplinary Research*, 2024.
[16] D. Joshi, "Detecting regime changes in financial markets using hidden Markov models and directional changes," *Int. J. For Multidisciplinary Research*, 2023.
[17] "Reinforcement-learning portfolio allocation with dynamic embedding and online meta-learning," *arXiv preprint arXiv:2501.17992*, 2024.
[18] "How machine learning is transforming portfolio optimization," *CFA Institute Blogs*, Sept. 2024.
[19] J. Behera et al., "Prediction based mean-value-at-risk portfolio optimization using machine learning regression models," *Engineering Applications of Artificial Intelligence*, vol. 117, 2023.
[20] J. Ghosn and Y. Bengio, "Multi-task learning for stock selection," in *Advances in Neural Information Processing Systems (NeurIPS)*, 1997.
[21] H. J. Park et al., "Stock market forecasting using a multi-task approach integrating long short-term memory and random forest," *Applied Soft Computing*, vol. 114, 2022.
[22] M. Arjovsky, "Out of distribution generalization in machine learning," *arXiv preprint arXiv:2103.02667*, Mar. 2021.
[23] K. Li et al., "Probing out-of-distribution generalization in machine learning," *Communications Engineering*, vol. 4, Jan. 2025.
[24] X. Zhang et al., "Deep stable learning for out-of-distribution generalization," in *Proc. IEEE/CVF Conf. Computer Vision Pattern Recognition (CVPR)*, 2021.
[25] "Transfer learning for financial data predictions: A systematic review," *arXiv preprint arXiv:2409.17183*, Sept. 2024.
[26] "Transfer learning in crisis prediction," *Meegle AI Topics*, Oct. 2025.
[27] "Early warning indicators of banking crises," *Bank of Finland Bulletin*, June 2015.
[28] "Network-based early warning system to predict financial crises," *SSRN Working Paper*, Jan. 2019.
[29] "Continual deep reinforcement learning for financial trading using bidirectional knowledge transfer," *SSRN Working Paper 4907420*, July 2024.
[30] "Continual learning and catastrophic forgetting," *University of Illinois Chicago*, 2022.
[31] P. García-Santaclara et al., "Overcoming catastrophic forgetting in tabular data continual learning," *Knowledge-Based Systems*, vol. 307, 2025.
[32] Z. Liu et al., "VIX and stock market volatility predictability: A new approach," *J. Empirical Finance*, vol. 69, pp. 93–110, 2022.
[33] "VIX spike time series forecasting," *SSRN Working Paper 5106657*, Dec. 2024.
[34] "A novel drift detection algorithm for machine learning in trading," *QuantInsti Blog*, Mar. 2025.
[35] "A generative approach for simulating concept drift in financial markets," *arXiv preprint arXiv:2509.11844*, Aug. 2025.
[36] H. Pabuccu et al., "Feature selection with annealing for forecasting financial time series," *arXiv preprint arXiv:2303.02223*, Mar. 2023.

[37] "Ensemble machine learning application and feature importance analysis for stock price prediction," *American J. Pure Applied Sciences*, vol. 6, no. 6, Nov. 2025.

[38] B. Dou et al., "Machine learning methods for small data challenges in molecular science," *Chemical Reviews*, vol. 123, no. 13, pp. 8736–8780, June 2023.

[39] P. Xu et al., "Small data machine learning in materials science," *npj Computational Materials*, vol. 9, no. 42, Mar. 2023.

[40] "FinRL-Meta: Market environments and benchmarks for data-driven financial reinforcement learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[41] S. Park et al., "Fast adaptation to super-resolution networks via meta-learning," in *Proc. European Conf. Computer Vision (ECCV)*, 2020.

[42] C. Min et al., "Meta-learning for adaptation of deep optical flow networks," in *Proc. IEEE/CVF Winter Conf. Applications Computer Vision (WACV)*, 2023.

[43] "Meta-learning with personalized learning rates for rapid task adaptation," *OpenReview*, Feb. 2024.